CURRICULUM VITAE

Muhammad Abu Bakar

Geospatial Data Specialist

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Professional experience

11/2025 – present

Geospatial Data Specialist · Beagle Systems, Hamburg

Beagle Systems flies long-range drones beyond visual line of sight (BVLOS) to monitor pipelines, utilities and other infrastructure.

- Developed an open-source drone LiDAR pipeline that georeferences point clouds with PPK and classifies them into ground, vegetation, buildings, conductors and towers. It rebuilds conductors where the scan has gaps by modeling their sag, rates vegetation near the wires by risk category, and is now being extended to tree-fall risk analysis.
- Built an AI-driven right-of-way and construction-site monitoring pipeline that georeferences drone images through deep feature matching and flags objects of different categories along pipelines. A fine-tuned foundation model detects the objects, an open-source VLM verifies each one, and an NVIDIA Jetson runs detection in real time.
- Fine-tuned a foundation model for building segmentation, using training labels created in-house with colleagues from 10 cm orthophotos. Scaled it to city level across the Cologne area, mapping buildings directly from the raw imagery.
- Built a geospatial backend with PostGIS and TiTiler that serves large GeoTIFFs to the Beagle Systems frontend as dynamic map tiles. Supported flight operations with pre-flight primary and duplicate checks, and helped expand the flight-planning geodatabase with new data sources.

03/2024 – 10/2025

Geospatial (EO) Data Scientist, part-time · Karuna Technology UG, Germany

- Differential InSAR pipeline with ISCE: automated acquisition, preprocessing and interferogram generation.
- Google Earth Engine workflows for Sentinel-1 time series (Refined Lee filter) feeding a soil-moisture ML model.
- Urban tree-species classification from fused multispectral and SAR data; training data for tree-crown segmentation.
- Vector data pre- and post-processing in QGIS.

04/2021 – 02/2024

GIS & RS Specialist · MapTransfer, Germany

- Spatial data workflows for a construction company: georeferencing, orthophoto mosaics, cloud-optimized GeoTIFFs on AWS, PostGIS–QGIS integration.
- In-house courses on LiDAR in QGIS and spatial data analysis; course development for Spatial SQL, ALKIS and GeoServer.
- Geodata management and visualization for a telecom project based on ALKIS.
- Web GIS: PostGIS vector layers in OpenLayers.

Education

2017 – 2020

MSc Geology · University of Freiburg, Germany

- Thesis: Simulation of multispectral and hyperspectral imaging techniques for mapping of epithermal ore deposits in vegetated zones.

2016 – 2020

MSc Remote Sensing & Geographic Information System · National University of Sciences and Technology, Pakistan

- Thesis: Development of an open-source Python-based slope stability model that incorporates remote sensing and geotechnical data for slope failure assessment.

2012 – 2015

BSc Geology · Bahria University, Pakistan

- Thesis: Petrophysical analysis of Joya Mair – Well 4.

Academic projects

Landsat · Sentinel-2	Glacier mass balance of Mýrdalsjökull from historical imagery and DEMs: the melting rate doubled after 2000.
MODIS · TRMM	Snow cover and precipitation variability, 2000–2015.
Landsat 7	Snow-cover change in the Kabul Basin and its impact on water resources.
Landsat 8 · ASTER	Land cover, erosion and morphometry of Mauna Kea.

Further training

2022

NASA Applied Remote Sensing Training (ARSET): five trainings on SAR-based disaster assessment, flood monitoring and modeling, crop mapping with polarimetric SAR and optical data, air-quality model output, and pre- and post-fire monitoring.

UNAVCO / GAGE short course: InSAR processing and time-series analysis for geophysical applications (ISCE, ARIA-tools, MintPy).

Skills

GIS	ArcGIS Pro, QGIS, ArcMap
Remote sensing	Multispectral, hyperspectral and SAR data; ENVI, ERDAS Imagine, eCognition, ESA SNAP
LiDAR	CloudCompare, WhiteboxTools; DEM and DSM generation
Cloud	Google Earth Engine, AWS (EC2, S3)
Databases	PostgreSQL/PostGIS, SpatiaLite, DuckDB, SQL
Programming	Python (GDAL, Fiona, Rasterio, scikit-learn, TensorFlow, Cartopy), R, Spatial SQL, Bash
Machine learning	CatBoost, CNNs, fine-tuned foundation models, VLMs
Web mapping	Leaflet, OpenLayers, GeoServer, TiTiler
Other	Git, LaTeX, Microsoft Office

Languages

English	C2
German	B2

Freiburg im Breisgau, 14 October 2026

Signature